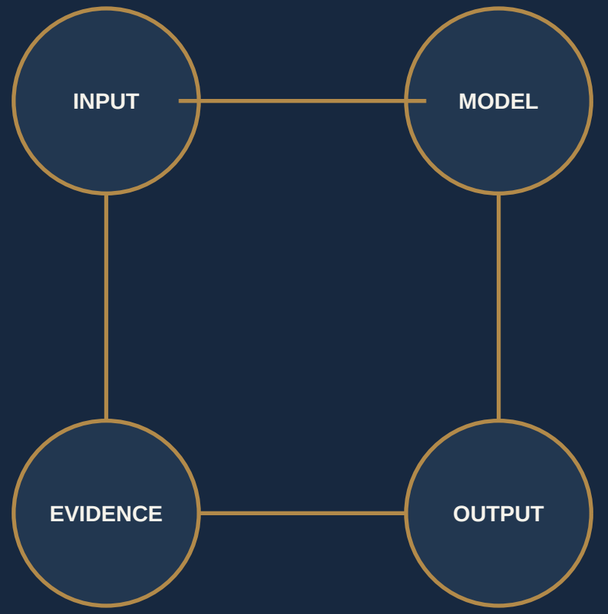
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Computer Graphics

AI.112 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define 2d/3d graphics.
- Apply key methods including transformations.
- Build or analyze a practical system involving coordinate systems.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

2D/3D graphics

02

METHODS

Transformations

03

SYSTEMS

Coordinate systems

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · 2D/3D graphics
2D/3D graphics

02 · Transformations
Transformations

03 · Coordinate systems
Coordinate systems

04 · Rendering
Rendering

05 · Lighting
Lighting

06 · Textures
Textures

07 · Cameras
Cameras

08 · Animation and introductory rea
Animation and introductory real-time graphics

Core ideas, made explicit

AI.112 • Distinguish definitions, assumptions and evidence

- 2D/3D graphics
- Transformations
- Coordinate systems
- Rendering
- Lighting

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.112 • Distinguish definitions, assumptions and evidence

- Textures
- Cameras
- Animation and introductory real-time graphics
- Mandatory practicals: Create 2D and 3D scenes
- Implement transformations and build an interactive graphical application

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
2D/3D graphics

INPUTS
Transformations

METHOD
Coordinate systems

OUTPUT
Rendering

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving 2d/3d graphics.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using 2d/3d graphics.
- Implement or simulate a baseline using transformations.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.